EFY TESTED
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# HIGH/LOW-VOLTAGE CUT OUT
## With Built-in Delay

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Most electronic and electric devices require a voltage within certain limits. If the voltage applied is higher than normal, the current drawn by the circuit may exceed the rated value and damage the circuit. If the voltage is too low, the device may not function properly. In the case of a refrigerator or an air-conditioner, the compressor may not work and could lead to burning of the motor. Additionally, all devices that use compressors, such as refrigerators and air-conditioners, should be restarted after a gap of two to three minutes. Otherwise, the compressor, which is full of refrigerant, may not be able to rotate and may get burnt out by drawing excessive current.

The circuit described below cuts out power to the load if the input voltage is too high or too low. The voltage at which the power is cut off can be adjusted. If the power fails briefly and resumes, it switches on power to the load after a gap of a few minutes; this time can be adjusted. If the voltage becomes too high or too low and then normalises, the device switches on after a gap of a few minutes. Fig. 1 shows the EFY lab's prototype on a breadboard.

### Circuit and working

The circuit diagram of the voltage cut out with delay is shown in Fig.

| PARTS LIST | |
|---|---|
| *Semiconductors:* | |
| IC1 | - LM324, op-amp |
| T1 | - 2N3904 NPN transistor |
| D1-D4 | - 1N4007 rectifier diode |
| LED1, LED2 | - 5mm LED (green, red) |
| ZD1 | - 6V zener diode |
| *Resistors (all 1/4-watt, ±5% carbon):* | |
| R1, R7, R8, R10, R11 | - 1-kilo-ohm |
| R2 | - 4.7-kilo-ohm |
| R3, R9 | - 10-kilo-ohm |
| R4, R6 | - 1-mega-ohm |
| R5 | - 2.2-kilo-ohm |
| VR1, VR2 | - 10-kilo-ohm preset |
| VR3 | - 5-kilo-ohm preset |
| *Capacitors:* | |
| C1 | - 100µF, 35V electrolytic |
| *Miscellaneous:* | |
| RL1 | - 12V SPDT relay |
| CON1-CON3 | - 2-pin connector |
| | - 15V unregulated power supply |
| | - 230V AC appliance as load |

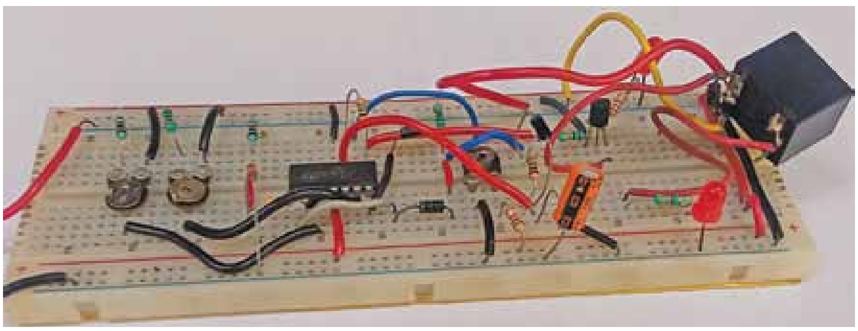

Fig. 1: EFY lab's prototype

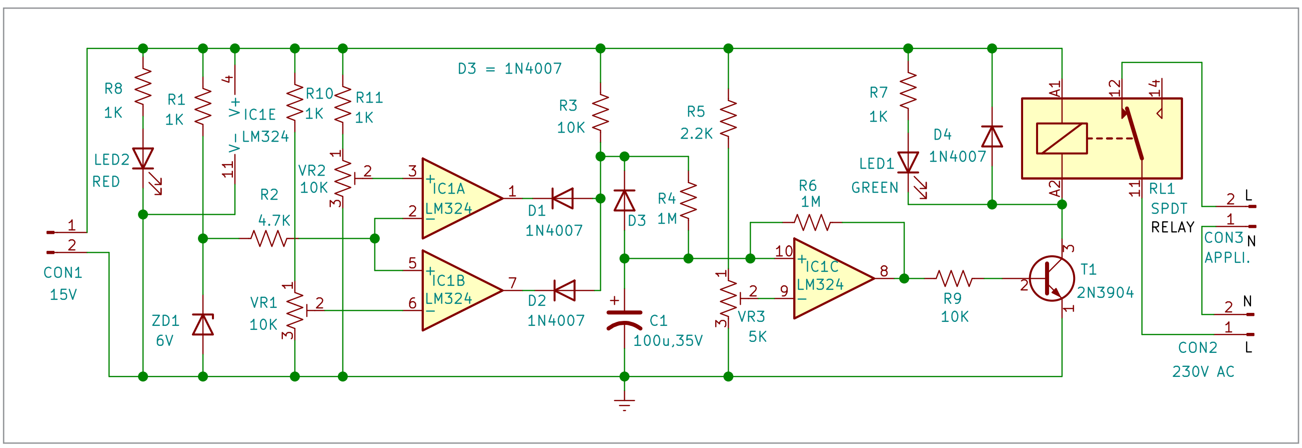


Fig. 2: Circuit diagram